Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
8/2	(a)	(i)		either of following • (reaction) temperature above melting point of iron • melting point of iron below reaction temperature / 2500°C			1	1		1
		(ii)		Al_2O_3 (1) 2 Fe (1) product **must** be correct for balancing mark to be awarded		2		2	1	
		(iii)		aluminium is oxidised because it gains oxygen do not accept aluminium oxide is oxidised accept 'aluminium is oxidised because it loses electrons'	1			1		
		(iv)		magnesium aluminium iron **must** be in correct order			1	1		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(b)	(i)		What are the positions of the four metals in the reactivity series? ✓			1	1		1
		(ii)		**D**		1		1		1
		(iii)		any of following for (1) • copper in copper(II) sulfate • tin in tin(II) sulfate • iron in iron(II) sulfate • zinc in zinc sulfate • metal in its own sulfate solution • metals in their own sulfate solutions metals do not displace themselves from solution / metals do not react with their own sulfate (1)	2			2		2
	(c)	(i)		any of following • silvery/grey solid formed • (brown) copper turns silvery/grey • (colourless) solution turns blue neutral answer – 'metal changes colour' or 'solution changes colour'	1			1		1
		(ii)		$Cu + 2AgNO_3 \rightarrow Cu(NO_3)_2 + 2Ag$ products (1) balancing (1) reactants and products **must** be correct for balancing mark to be awarded		2		2		2
				Question 8/2 total	**4**	**5**	**3**	**12**	**1**	**8**